

OCNA SUGATAG, Romania

WMO: 150150

Lat: 47.7769N

Lon: 23.9406E

Elev: 1655

StdP: 13.84

Time Zone: 2.00 (EUE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	12.9	-1.8	5.3	12.3	3.9	7.1	16.0	14.1	36.8	11.7	35.8	2.6	190	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.1	86.0	67.9	82.8	66.9	79.9	65.3	70.1	81.5	68.3	79.8	66.7	77.4	4.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	101.4	76.3	64.3	96.0	74.0	62.8	91.1	72.0	35.1	81.5	33.6	79.9	32.3	77.5	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.7	9.7	8.1	DB	3.3	90.1	4.6	3.0	0.0	92.3	-2.7	94.0	-5.3	95.7	-8.6	97.9
			WB	2.3	72.7	4.5	1.3	-0.9	73.6	-3.5	74.4	-6.0	75.1	-9.2	76.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	28.6	32.2	39.5	50.6	57.8	63.8	66.3	66.8	58.3	49.6	41.5	31.7	(6)
	DBStd	15.20	8.46	8.58	8.30	6.69	6.24	5.79	5.29	5.54	6.76	7.13	8.35	8.12	(7)
	HDD50	2528	662	499	334	72	11	0	0	0	11	97	274	568	(8)
	HDD65	6112	1127	919	791	432	232	89	48	45	215	477	704	1033	(9)
	CDD50	2163	0	0	8	91	254	414	506	522	262	86	20	0	(10)
	CDD65	272	0	0	0	1	10	53	90	102	15	1	0	0	(11)
	CDH74	2582	0	0	0	20	130	464	794	992	178	4	0	0	(12)
	CDH80	640	0	0	0	1	7	80	213	305	34	0	0	0	(13)
	WSAvg	3.2	2.5	2.7	3.9	3.9	3.5	3.4	3.4	3.2	3.1	2.9	3.0	2.5	(14)
	PrecAvg	33.8	2.0	1.9	2.2	2.3	3.5	3.9	4.1	3.1	3.1	2.8	2.4	2.4	(15)
(16) Precipitation	PrecMax	43.9	4.8	5.7	6.0	3.6	7.8	6.9	9.0	9.3	6.1	6.6	5.3	4.2	(16)
	PrecMin	25.7	0.9	0.7	0.4	0.8	1.0	1.4	0.8	0.7	0.7	0.3	0.2	0.4	(17)
	PrecStd	5.5	0.9	1.0	1.5	0.8	1.4	1.6	2.3	1.7	1.4	1.7	1.2	1.1	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	55.3	64.3	77.0	81.0	85.6	89.1	90.8	84.2	73.6	65.6	53.3	(19)
		MCWB	44.0	46.2	50.1	57.6	61.8	67.9	68.5	67.4	65.0	58.5	54.5	46.3	(20)
	2%	DB	45.3	49.8	59.5	70.8	77.7	82.3	85.5	86.9	78.7	69.2	60.3	49.8	(21)
		MCWB	42.0	42.5	47.6	54.1	60.9	66.8	68.4	67.7	62.4	57.0	51.4	44.9	(22)
	5%	DB	42.8	46.3	55.5	67.3	74.3	79.3	81.8	83.2	74.7	65.0	56.5	46.2	(23)
		MCWB	40.1	41.0	45.4	52.2	60.2	65.4	67.2	67.1	61.1	55.2	49.7	42.3	(24)
	10%	DB	39.6	43.2	51.7	63.5	70.5	76.2	78.3	79.7	70.7	61.1	53.3	42.8	(25)
		MCWB	37.3	39.2	43.7	50.6	58.4	63.8	65.6	66.0	59.6	53.3	47.8	39.8	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	46.6	51.5	58.7	64.9	71.3	72.2	72.5	66.3	60.2	55.6	48.2	(27)
		MCDB	47.3	53.5	61.2	73.0	76.5	82.8	83.7	84.8	80.2	69.8	62.2	51.6	(28)
	2%	WB	42.6	43.9	48.8	56.0	62.8	68.8	69.9	70.2	64.1	58.1	52.8	45.1	(29)
		MCDB	44.8	48.5	57.6	69.0	73.8	79.4	81.5	81.7	75.2	66.8	58.6	49.3	(30)
	5%	WB	40.4	41.9	46.5	53.6	61.0	66.8	68.2	68.2	62.4	56.3	50.5	42.6	(31)
		MCDB	42.5	45.8	53.9	64.3	71.8	76.7	79.8	79.7	72.5	63.5	55.3	46.2	(32)
	10%	WB	37.5	39.7	44.3	51.5	59.4	64.9	66.5	66.5	60.4	54.2	48.3	39.9	(33)
		MCDB	39.4	43.0	50.6	60.9	69.1	73.8	76.8	77.5	69.2	60.3	53.1	42.6	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.1	15.3	18.6	19.3	19.5	19.9	21.1	18.9	17.4	13.0	9.4	(35)
		MCDBR	11.3	15.6	21.6	24.6	25.2	24.2	25.2	26.8	25.3	22.8	18.1	13.4	(36)
		MCWBR	8.4	10.1	12.0	11.7	12.0	11.3	11.3	11.0	11.6	12.1	10.4	9.5	(37)
	5% WB	MCDBR	9.9	14.2	19.0	21.7	22.8	22.3	23.3	23.9	22.6	20.1	15.6	12.7	(38)
		MCWBR	7.6	9.5	10.9	10.9	11.3	11.2	11.1	11.0	11.2	11.0	9.3	9.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.311	0.340	0.387	0.441	0.424	0.419	0.442	0.443	0.402	0.370	0.342	0.313	(40)
	taud		2.254	2.191	2.149	2.081	2.200	2.247	2.194	2.188	2.261	2.322	2.333	2.313	(41)
	Ebn at Noon		236	254	259	256	267	268	260	253	252	242	224	221	(42)
	Edh at Noon		25	33	40	48	44	42	44	43	36	29	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg		341	600	966	1362	1668	1795	1770	1628	1144	744	407	257	(44)
	RadStd		52	82	136	141	173	130	162	150	150	102	58	38	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+1.26	N/A	N/A	+1.87	N/A	N/A	-270	-389	+189	+70

Nomenclature: See separate page